

Appendix 3.

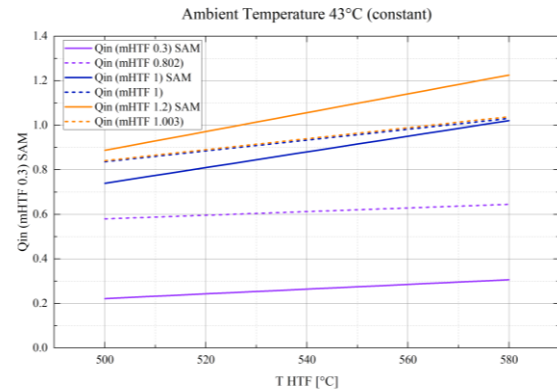
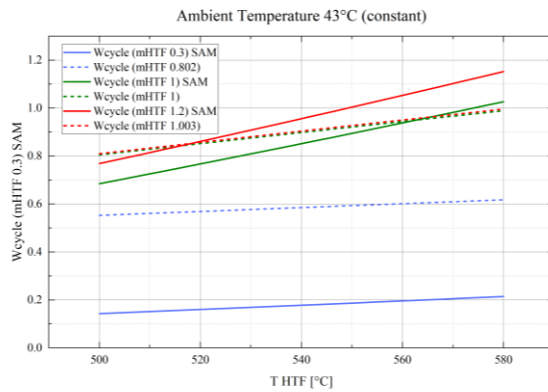
Parametric Results for Power Block Characterization

This appendix presents a set of parametric results derived from the thermo-energetic modeling of the power block in EES, based on the formulation proposed by Wagner (2008) [12] and Wagner & Gilman (2011) [66]. The main objective is to characterize the behavior of the Rankine cycle through simulations performed in the System Advisor Model (SAM) environment, considering systematic variations in thermal fluid (HTF) temperature and mass flow rate, as well as ambient temperature.

The graphs illustrate the behavior of the net power output of the cycle (Wcycle) and the thermal input required by the power block (Qin), both normalized with respect to reference conditions. The simulation scenarios include:

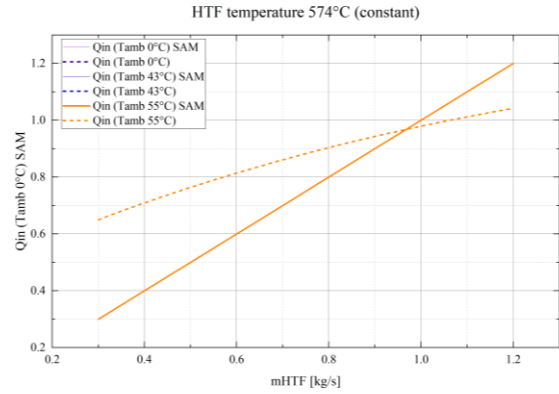
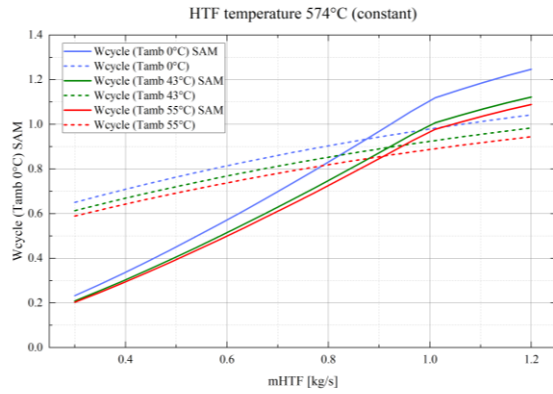
- Constant ambient temperature (43 °C) with variation in THTF
- Constant HTF temperature (574 °C) with variation in mHTF
- Constant mass flow rate (mHTF = 1 kg/s) with variation in ambient temperature

The corresponding numerical tables generated from the model developed in this study are included alongside the graphs.



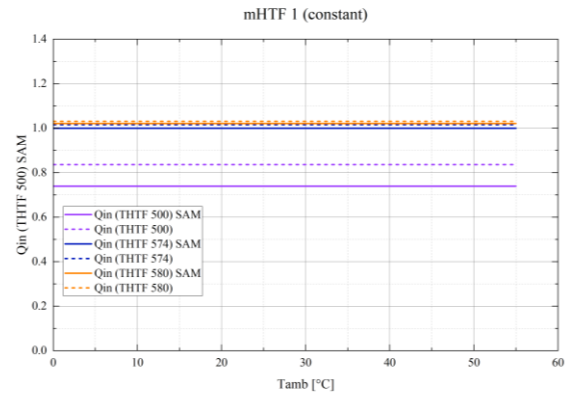
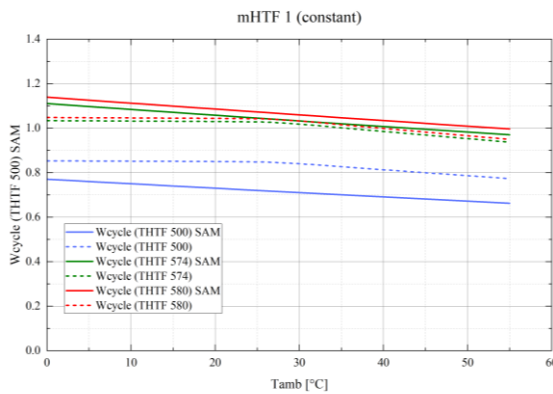
T HTF [°C]	Wcycle (m	Wcycle (m	Wcycle (m	Wcycle (m	Wcycle (m	Wcycle (m
500	0.14323	0.553	0.68515	0.8052	0.76894	0.81
504.211	0.14681	0.5564	0.70225	0.8151	0.78813	0.8199
508.421	0.1504	0.5598	0.71945	0.8249	0.80744	0.8299
512.632	0.15402	0.5632	0.73676	0.8347	0.82686	0.8399
516.842	0.15766	0.5666	0.75417	0.8446	0.8464	0.8498
521.053	0.16132	0.57	0.77168	0.8544	0.86605	0.8597
525.263	0.165	0.5734	0.78928	0.8641	0.88581	0.8696
529.474	0.16871	0.5768	0.80699	0.8739	0.90569	0.8795
533.684	0.17243	0.5802	0.8248	0.8836	0.92567	0.8894
537.895	0.17617	0.5836	0.8427	0.8933	0.94576	0.8992
542.105	0.17993	0.587	0.86069	0.903	0.96596	0.909
546.316	0.18371	0.5904	0.87878	0.9127	0.98626	0.9188
550.526	0.18751	0.5938	0.89697	0.9223	1.00666	0.9286
554.737	0.19134	0.5972	0.91524	0.9319	1.02718	0.9383
558.947	0.19518	0.6006	0.93361	0.9415	1.04779	0.948
563.158	0.19903	0.604	0.95207	0.9511	1.0685	0.9577
567.368	0.20291	0.6075	0.97061	0.9606	1.08932	0.9673
571.579	0.20681	0.6109	0.98925	0.9701	1.11023	0.9769
575.789	0.21072	0.6143	1.00797	0.9796	1.13124	0.9865
580	0.21465	0.6177	1.02677	0.9891	1.15235	0.9961

T HTF [°C]	Qin (mHTF	Qin (mHTF	Qin (mHTF	Qin (mHTF	Qin (mHTF	Qin (mHTF
500	0.22183	0.5797	0.73944	0.8362	0.88732	0.8411
504.211	0.22628	0.5831	0.75426	0.8464	0.90512	0.8514
508.421	0.23073	0.5865	0.76909	0.8566	0.92291	0.8618
512.632	0.23517	0.5899	0.78391	0.8668	0.9407	0.8721
516.842	0.23962	0.5934	0.79874	0.877	0.95849	0.8825
521.053	0.24407	0.5968	0.81357	0.8872	0.97628	0.8929
525.263	0.24852	0.6002	0.82839	0.8975	0.99407	0.9032
529.474	0.25297	0.6037	0.84322	0.9077	1.01186	0.9136
533.684	0.25741	0.6071	0.85804	0.9179	1.02965	0.924
537.895	0.26186	0.6105	0.87287	0.9282	1.04744	0.9344
542.105	0.26631	0.614	0.8877	0.9384	1.06523	0.9448
546.316	0.27076	0.6174	0.90252	0.9487	1.08302	0.9552
550.526	0.2752	0.6209	0.91735	0.9589	1.10082	0.9655
554.737	0.27965	0.6243	0.93217	0.9692	1.11861	0.9759
558.947	0.2841	0.6278	0.947	0.9794	1.1364	0.9863
563.158	0.28855	0.6312	0.96182	0.9897	1.15419	0.9968
567.368	0.293	0.6347	0.97665	0.9999	1.17198	1.007
571.579	0.29744	0.6381	0.99148	1.01	1.18977	1.018
575.789	0.30189	0.6416	1.0063	1.02	1.20756	1.028
580	0.30634	0.645	1.02113	1.031	1.22535	1.038



mHTF [kg/s]	Wcycle (Ta	Wcycle (Ta	Wcycle (Ta	Wcycle (Ta	Wcycle (Ta	Wcycle (Ta
0.3	0.23219	0.6501	0.20905	0.6135	0.20288	0.5892
0.34737	0.28094	0.6787	0.25295	0.6405	0.24548	0.6151
0.39474	0.33173	0.7062	0.29888	0.6664	0.28986	0.64
0.44211	0.38439	0.7326	0.34609	0.6914	0.33587	0.664
0.48947	0.43877	0.7582	0.39505	0.7155	0.38339	0.6871
0.53684	0.49475	0.7828	0.44545	0.7387	0.4323	0.7094
0.58421	0.55224	0.8065	0.49721	0.7611	0.48253	0.7309
0.63158	0.61114	0.8294	0.55025	0.7827	0.534	0.7517
0.67895	0.67139	0.8514	0.60449	0.8035	0.58664	0.7716
0.72632	0.73291	0.8726	0.65988	0.8234	0.6404	0.7908
0.77368	0.79564	0.8929	0.71636	0.8426	0.69522	0.8092
0.82105	0.85954	0.9124	0.7739	0.861	0.75105	0.8269
0.86842	0.92456	0.9312	0.83243	0.8787	0.80786	0.8439
0.91579	0.99065	0.9491	0.89194	0.8957	0.86561	0.8602
0.96316	1.05777	0.9663	0.95237	0.9119	0.92426	0.8758
1.01053	1.1188	0.9828	1.00732	0.9275	0.97759	0.8907
1.05789	1.15414	0.9986	1.03914	0.9424	1.00846	0.905
1.10526	1.1873	1.0137	1.06899	0.9567	1.03744	0.9187
1.15263	1.21814	1.0283	1.09676	0.9703	1.06439	0.9319
1.2	1.2465	1.0422	1.1223	0.9835	1.08917	0.9445

mHTF [kg/s]	Qin (Tamb	Qin (Tamb	Qin (Tamb	Qin (Tamb	Qin (Tamb	Qin (Tamb
0.3	0.3	0.6501	0.3	0.6501	0.3	0.6501
0.34737	0.34737	0.6787	0.34737	0.6787	0.34737	0.6787
0.39474	0.39474	0.7062	0.39474	0.7062	0.39474	0.7062
0.44211	0.44211	0.7326	0.44211	0.7326	0.44211	0.7326
0.48947	0.48947	0.7582	0.48947	0.7582	0.48947	0.7582
0.53684	0.53684	0.7828	0.53684	0.7828	0.53684	0.7828
0.58421	0.58421	0.8065	0.58421	0.8065	0.58421	0.8065
0.63158	0.63158	0.8294	0.63158	0.8294	0.63158	0.8294
0.67895	0.67895	0.8514	0.67895	0.8514	0.67895	0.8514
0.72632	0.72632	0.8726	0.72632	0.8726	0.72632	0.8726
0.77368	0.77368	0.8929	0.77368	0.8929	0.77368	0.8929
0.82105	0.82105	0.9124	0.82105	0.9124	0.82105	0.9124
0.86842	0.86842	0.9312	0.86842	0.9312	0.86842	0.9312
0.91579	0.91579	0.9491	0.91579	0.9491	0.91579	0.9491
0.96316	0.96316	0.9663	0.96316	0.9663	0.96316	0.9663
1.01053	1.01053	0.9828	1.01053	0.9828	1.01053	0.9828
1.05789	1.05789	0.9986	1.05789	0.9986	1.05789	0.9986
1.10526	1.10526	1.0137	1.10526	1.0137	1.10526	1.0137
1.15263	1.15263	1.0283	1.15263	1.0283	1.15263	1.0283
1.2	1.2	1.0422	1.2	1.0422	1.2	1.0422



mHTF [kg/s]	Qin (Tamb	Qin (Tamb	Qin (Tamb	Qin (Tamb	Qin (Tamb	Qin (Tamb
0.3	0.3	0.6501	0.3	0.6501	0.3	0.6501
0.34737	0.34737	0.6787	0.34737	0.6787	0.34737	0.6787
0.39474	0.39474	0.7062	0.39474	0.7062	0.39474	0.7062
0.44211	0.44211	0.7326	0.44211	0.7326	0.44211	0.7326
0.48947	0.48947	0.7582	0.48947	0.7582	0.48947	0.7582
0.53684	0.53684	0.7828	0.53684	0.7828	0.53684	0.7828
0.58421	0.58421	0.8065	0.58421	0.8065	0.58421	0.8065
0.63158	0.63158	0.8294	0.63158	0.8294	0.63158	0.8294
0.67895	0.67895	0.8514	0.67895	0.8514	0.67895	0.8514
0.72632	0.72632	0.8726	0.72632	0.8726	0.72632	0.8726
0.77368	0.77368	0.8929	0.77368	0.8929	0.77368	0.8929
0.82105	0.82105	0.9124	0.82105	0.9124	0.82105	0.9124
0.86842	0.86842	0.9312	0.86842	0.9312	0.86842	0.9312
0.91579	0.91579	0.9491	0.91579	0.9491	0.91579	0.9491
0.96316	0.96316	0.9663	0.96316	0.9663	0.96316	0.9663
1.01053	1.01053	0.9828	1.01053	0.9828	1.01053	0.9828
1.05789	1.05789	0.9986	1.05789	0.9986	1.05789	0.9986
1.10526	1.10526	1.0137	1.10526	1.0137	1.10526	1.0137
1.15263	1.15263	1.0283	1.15263	1.0283	1.15263	1.0283
1.2	1.2	1.0422	1.2	1.0422	1.2	1.0422

mHTF [kg/s]	Qin (Tamb	Qin (Tamb	Qin (Tamb	Qin (Tamb	Qin (Tamb	Qin (Tamb
0.3	0.3	0.6501	0.3	0.6501	0.3	0.6501
0.34737	0.34737	0.6787	0.34737	0.6787	0.34737	0.6787
0.39474	0.39474	0.7062	0.39474	0.7062	0.39474	0.7062
0.44211	0.44211	0.7326	0.44211	0.7326	0.44211	0.7326
0.48947	0.48947	0.7582	0.48947	0.7582	0.48947	0.7582
0.53684	0.53684	0.7828	0.53684	0.7828	0.53684	0.7828
0.58421	0.58421	0.8065	0.58421	0.8065	0.58421	0.8065
0.63158	0.63158	0.8294	0.63158	0.8294	0.63158	0.8294
0.67895	0.67895	0.8514	0.67895	0.8514	0.67895	0.8514
0.72632	0.72632	0.8726	0.72632	0.8726	0.72632	0.8726
0.77368	0.77368	0.8929	0.77368	0.8929	0.77368	0.8929
0.82105	0.82105	0.9124	0.82105	0.9124	0.82105	0.9124
0.86842	0.86842	0.9312	0.86842	0.9312	0.86842	0.9312
0.91579	0.91579	0.9491	0.91579	0.9491	0.91579	0.9491
0.96316	0.96316	0.9663	0.96316	0.9663	0.96316	0.9663
1.01053	1.01053	0.9828	1.01053	0.9828	1.01053	0.9828
1.05789	1.05789	0.9986	1.05789	0.9986	1.05789	0.9986
1.10526	1.10526	1.0137	1.10526	1.0137	1.10526	1.0137
1.15263	1.15263	1.0283	1.15263	1.0283	1.15263	1.0283
1.2	1.2	1.0422	1.2	1.0422	1.2	1.0422